

Four-Step Gap Rigidity for the Toeplitz Commutant of $T_{z^2+\bar{z}^2}$

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Status. Substantial partial result, likely valid, pending human review. The theorem removes the source paper’s global Fourier cutoff for two natural classes of symbols with genuinely infinite angular spectra. It does not settle the unrestricted bounded-symbol problem.

1 Source question

Let $A^2(\mathbb{D})$ be the Bergman space of the unit disk and let T_f denote the Toeplitz operator with bounded symbol f . Axler, Čučković and Rao asked whether, for every bounded harmonic g which is neither analytic nor anti-analytic,

$$[T_f, T_g] = 0 \implies f = \alpha g + \beta$$

for every bounded f . Alsabi and Louhichi quote the question on PDF page 2 of [1]:

the identity, T_g commutes with T_f .

The next natural step was to relax the hypothesis of the previous theorem in order to obtain results for a larger class of symbols. In [2], S. Axler, Ž. Čučković, and N. V. Rao proved that analytic Toeplitz operators commute only with other such operators. Their result can be stated as follows:

Theorem 2 (Axler, Čučković & Rao). *If g is a nonconstant analytic function in $\mathbb{D}$ and if f is bounded in $\mathbb{D}$ such that $T_f T_g = T_g T_f$, then f must be analytic too.*

For Theorem 2, the authors do not ask the function f to be harmonic but only bounded. However this was not without cost. In fact the hypothesis on the symbol g is stronger than the one in Theorem 1 since here g has to be analytic. Finally, the authors conclude [2] by asking the following open problem: **“Suppose g is a bounded harmonic function in $\mathbb{D}$ that is neither analytic nor antianalytic. If f is a bounded function in $\mathbb{D}$ such that T_f and T_g commute, must f be of the form $\alpha g + \beta$ for some constants α, β ?”** The first partial answers to this problem can be found in [6] and [7].

2. QUASIHOMOGENEOUS TOEPLITZ OPERATORS

Definition 1. *A symbol f is said to be quasihomogeneous of order p , and the*

Figure 1: The open problem as quoted in [1, p. 2].

For the fixed symbol

$$h(z) = z^2 + \bar{z}^2,$$

their main theorem proves the desired conclusion when the polar Fourier expansion of f is right-terminating. The result below permits infinitely many positive and negative modes.

For $f \in L^\infty(\mathbb{D})$ write

$$f(re^{i\theta}) = \sum_{k \in \mathbb{Z}} e^{ik\theta} f_k(r) \quad \text{in } L^2(\mathbb{D}),$$

and let

$$S(f) = \{k \in \mathbb{Z} : f_k \neq 0 \text{ in } L^2([0, 1], r \, dr)\}.$$

In each residue class modulo 4, regard consecutive integers at distance 4 as adjacent. A *four-step support component* is a connected component of $S(f)$ in this graph.

Theorem 1 (Four-step gap rigidity). *Let $f \in L^\infty(\mathbb{D})$ and suppose*

$$[T_f, T_{z^2 + \bar{z}^2}] = 0.$$

Each of the following hypotheses implies that, almost everywhere on $\mathbb{D}$,

$$\boxed{f = \alpha(z^2 + \bar{z}^2) + \beta}$$

for some $\alpha, \beta \in \mathbb{C}$.

- (i) *Every four-step support component of f is finite.*
- (ii) *The angular Fourier series is Wiener-summable,*

$$\sum_{k \in \mathbb{Z}} \|f_k\|_\infty < \infty,$$

and, in every residue class $q \pmod{4}$, there are arbitrarily large $k \equiv q \pmod{4}$ for which $f_k = 0$.

Thus (i) covers sparse symbols with infinitely many modes in both directions, while (ii) needs gaps only in the positive direction. Equivalently, under the Wiener hypothesis a non-affine commuting symbol, if one exists, must have an eventually gapless positive ray in at least one residue class modulo 4.

2 Proof intuition

Rotations split every Toeplitz operator into homogeneous pieces. Commuting with $T_{z^2 + \bar{z}^2}$ couples the piece of angular degree k only to the piece of degree $k + 4$. A missing Fourier mode is therefore an exact cut in the recurrence, not merely a small error. Every support block between two such cuts is itself the symbol of a Toeplitz operator commuting with $T_{z^2 + \bar{z}^2}$.

Finite blocks are bounded right-terminating symbols, so the source theorem says that each block is affine in $z^2 + \bar{z}^2$. Under Wiener summability, the one possible block extending to $-\infty$ also converges in L^∞ and is right-terminating, so the same theorem applies. Finally, the residue-class support of an affine symbol is so small that only the constant block and the block containing both degrees -2 and 2 can remain.

3 Homogeneous commutator decomposition

Let U_t be the rotation unitary on $A^2(\mathbb{D})$. An operator X is homogeneous of degree k when

$$U_t X U_t^* = e^{ikt} X.$$

The weak-operator Fourier coefficient

$$\Phi_k(X) = \frac{1}{2\pi} \int_0^{2\pi} e^{-ikt} U_t X U_t^* dt$$

is a bounded homogeneous operator. Rotation covariance of Bergman Toeplitz operators gives

$$\Phi_k(T_f) = T_{e^{ik\theta} f_k(r)} =: T_k.$$

The two summands T_{z^2} and $T_{\bar{z}^2}$ have degrees 2 and -2 . Taking the degree $k+2$ coefficient of the zero commutator yields, for every $k \in \mathbb{Z}$,

$$[T_k, T_{z^2}] + [T_{k+4}, T_{\bar{z}^2}] = 0. \quad (1)$$

Lemma 1 (Block separation). *Let B be a four-step support component. Suppose the symbol*

$$f_B = \sum_{k \in B} e^{ik\theta} f_k(r)$$

defines a bounded Toeplitz operator, with the sum interpreted in L^∞ when B is infinite. Then

$$[T_{f_B}, T_{z^2 + \bar{z}^2}] = 0.$$

Proof. Inside B , equation (1) is exactly the corresponding homogeneous coefficient of $[T_{f_B}, T_{z^2 + \bar{z}^2}]$. At either endpoint, the adjacent degree outside B is a missing Fourier mode, hence its operator is zero, and (1) again gives the endpoint coefficient. All other homogeneous coefficients vanish trivially. If B is infinite and the stated series converges in L^∞ , the operator series converges in norm, so the same coefficient argument passes to the sum. Therefore every homogeneous coefficient of the block commutator is zero. A bounded operator whose weak rotation Fourier coefficients all vanish is zero, proving the claim. $\square$

4 Proof of Theorem 1

We use the main theorem of Alsabi and Louhichi [1, Section 3]: if a bounded symbol u is right-terminating and T_u commutes with $T_{z^2 + \bar{z}^2}$, then

$$u = c(z^2 + \bar{z}^2) + c_0 \quad (2)$$

almost everywhere.

Assume first (i). For every support component B , the sum defining f_B is finite, hence $f_B \in L^\infty$, and Lemma 1 shows that its Toeplitz operator commutes with $T_{z^2 + \bar{z}^2}$. It is right-terminating, so (2) applies.

But B lies in one residue class modulo 4, whereas the only Fourier degrees in (2) are $-2, 0, 2$. A block in residue class 1 or 3 must therefore be zero. A block in residue class 0 must be zero unless it contains degree 0, in which case it is constant. A block in residue class 2 must be zero

unless it contains both adjacent degrees -2 and 2 , in which case it is a scalar multiple of $z^2 + \bar{z}^2$. Consequently every Fourier coefficient of f other than these three vanishes, and uniqueness of the L^2 Fourier expansion proves the claimed formula.

Now assume (ii). In a fixed residue class, the hypothesis of arbitrarily large missing indices implies that every support component is bounded above. All such components are finite except possibly one component extending to $-\infty$. Finite components are handled as above. For a lower-infinite component B , the Wiener assumption gives

$$\sum_{k \in B} \|e^{ik\theta} f_k(r)\|_\infty < \infty.$$

Hence f_B converges in L^∞ , is bounded and right-terminating, and Lemma 1 and (2) apply. The same residue-support argument eliminates every block except the possible constant and $z^2 + \bar{z}^2$ blocks. This proves (ii). $\square$

5 Verification, limitations, and upgrade audit

The proof is analytic and has no computational dependency. The two points most worth checking are the indices in (1) and the passage from a lower-infinite block to a bounded symbol in part (ii). Both are explicit: commutation degrees are $(k) + 2$ and $(k + 4) - 2$, and absolute convergence in L^∞ gives norm convergence of the corresponding Toeplitz operators.

The theorem does not settle the fixed-symbol problem for arbitrary bounded f . Eight focused upgrade attempts tested later literature, the Jacobi spectral commutant, compact commuting remainders, smooth boundary asymptotics, and quantitative tail decay before isolating the exact block mechanism and its Wiener upgrade. The remaining case contains an eventually gapless positive four-step tail. Replacing an exact gap by smallness would require a uniform stability estimate for the Mellin-transform recurrence; the source proof is qualitative and no such estimate was obtained.

6 Literature and novelty boundary

A bounded search on 17 August 2026 covered the run's cheap indexes, the exact source title and arXiv id, the citation neighborhood of the source and the original problem, Le and Tikaradze's later right-terminating theorem [2], and the 2024 finite quasihomogeneous-sum result [3]. Those located results retain right-termination or finite-sum restrictions. No explicit statement of Theorem 1 was found. Novelty confidence is moderate: the block argument is short once the source theorem and the four-step recurrence are placed together.

Human review is recommended, concentrating on Lemma 1, the support-component classification, and the precise hypotheses of the source right-terminating theorem.

References

- [1] H. Alsabi and I. Louhichi, *On the commutativity of a certain class of Toeplitz operators*, arXiv:1704.04757; published in *Operators and Matrices* 12 (2018), 863–879.
- [2] T. Le and A. Tikaradze, *Commutants of Toeplitz operators with harmonic symbols*, *New York J. Math.* 23 (2017), 1723–1731; arXiv:1706.01510.
- [3] H. Alsabi and I. Louhichi, *Commutants of the sum of two quasihomogeneous Toeplitz operators*, *Concr. Oper.* 11 (2024); arXiv:2409.14361.